



Shore

**Year 11
Mathematics
Yearly Examination
September 2013**

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Answer Questions 1–5 on the Multiple Choice Answer Sheet provided
- Start each of Questions 6–10 in a new writing booklet
- In Questions 6–10, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

Total marks – 80

Section I Pages 2–3

5 marks

- Attempt Questions 1–5
- Each question is worth 1 mark
- Allow about 10 minutes for this section

Section II Pages 4–8

75 marks

- Attempt Questions 6–10
- Each question is worth 15 marks
- Allow about 1 hour and 50 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

Section I

5 marks

Attempt Questions 1–5

Allow about 10 minutes for this section

Use the multiple choice answer sheet for Questions 1–5.

1 Which one of the following is equivalent to $\frac{x}{8} - \frac{x+1}{4}$?

- (A) $\frac{1-x}{8}$
- (B) $\frac{-1-x}{8}$
- (C) $\frac{2-x}{8}$
- (D) $\frac{-2-x}{8}$

2 Which one of the following is equal to $\frac{1+\sqrt{5}}{7-2\sqrt{5}}$?

- (A) $\frac{5\sqrt{5}-3}{29}$
- (B) $\frac{5\sqrt{5}-3}{39}$
- (C) $\frac{9\sqrt{5}+17}{29}$
- (D) $\frac{9\sqrt{5}+17}{39}$

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

- 3 The roots of a quadratic equation are -3 and 4 .

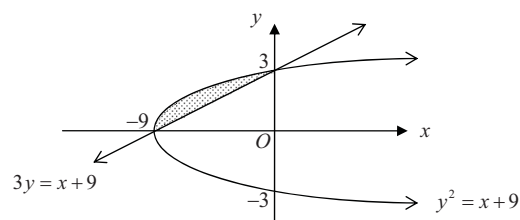
Which one of the following could be the quadratic equation?

- (A) $x^2 - x - 12 = 0$
 (B) $x^2 + x - 12 = 0$
 (C) $x^2 - x + 12 = 0$
 (D) $x^2 + x + 12 = 0$

- 4 What are the solutions to $\sqrt{3} \tan x = -1$ for $0^\circ \leq x \leq 360^\circ$?

- (A) 120° and 240°
 (B) 120° and 300°
 (C) 150° and 210°
 (D) 150° and 330°

- 5 The diagram shows the region enclosed by $3y = x + 9$ and $y^2 = x + 9$.



Which one of the following pairs of inequalities describes the shaded region in the diagram?

- (A) $3y \geq x + 9$ and $y^2 \geq x + 9$
 (B) $3y \geq x + 9$ and $y^2 \leq x + 9$
 (C) $3y \leq x + 9$ and $y^2 \geq x + 9$
 (D) $3y \leq x + 9$ and $y^2 \leq x + 9$

Section II

75 marks

Attempt Questions 6–10

Allow about 1 hour and 50 minutes for this section

Start each of Questions 6–10 in a new writing booklet.

In Questions 6–10, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks) Use a SEPARATE writing booklet.

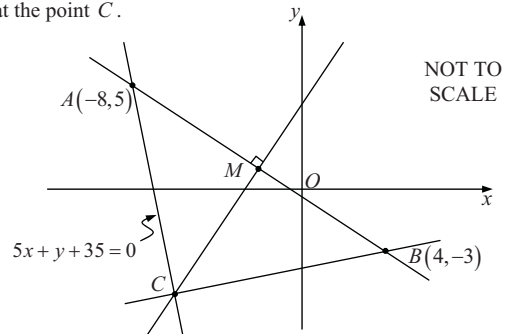
- (a) The selling price of a television is \$1056. This includes a 10% tax on the original price. Calculate the original price of the television. **2**
- (b) Find the perpendicular distance from the point $(-2, 5)$ to the line $4x + 3y = 12$. **2**
- (c) Factorise fully $x^2 - y^2 + 2x + 2y$. **2**
- (d) Solve $|3x - 1| < 7$. **2**
- (e) Find the value of m if the line $6x + my + 7 = 0$ is parallel to the line $3x + 2y - 4 = 0$. **2**
- (f) A regular polygon has an interior angle sum of 3960° . **2**
 How many sides does the polygon have?
- (g) A function is given by $f(x) = x^3 - 2x^2 + x$. **3**
 Find the values of x for which $f'(x) = 0$.

Question 7 (15 marks) Use a SEPARATE writing booklet.

(a) Differentiate $(1-4x^2)^5$ with respect to x .

2

(b) The diagram shows the points $A(-8,5)$ and $B(4,-3)$. The point M is the midpoint of the interval AB . The line $5x + y + 35 = 0$ meets the perpendicular bisector of AB at the point C .



(i) Find the coordinates of M .

1

(ii) Find the gradient of AB .

1

(iii) Show that the equation of the perpendicular bisector of AB is $3x - 2y + 8 = 0$.

2

(iv) Show that the coordinates of the point C are $(-6, -5)$.

2

(v) Prove that $\triangle ABC$ is right-angled at C .

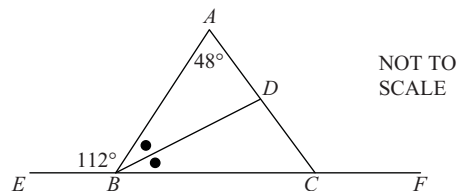
2

(vi) Find the area of $\triangle ABC$.

2

(c) In the diagram, $\angle ABE = 112^\circ$, $\angle BAC = 48^\circ$ and BD bisects $\angle ABC$.

3



Copy or trace the diagram into your writing booklet.

Find the size of $\angle BDC$, giving reasons for your answer.

- 5 -

Question 8 (15 marks) Use a SEPARATE writing booklet

(a) Simplify $\tan \theta \sqrt{1 - \sin^2 \theta}$.

2

(b) Find $f'(x)$ if $f(x) = \frac{1}{2x^4}$.

2

Write your answer in simplest form with positive integer indices.

(c) Find $\lim_{x \rightarrow 4} \left(\frac{x^2 + 3x - 4}{2x + 8} \right)$.

2

(d) Consider the curve $y = x^3 - 3x^2$.

3

Find the equation of the tangent at the point on the curve where $x = -1$.

(e) For what values of k does $3x^2 - 2kx + k = 0$ have real roots?

3

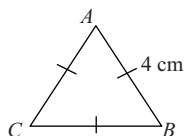
(f) Sketch the curve $y = \frac{1}{x-3} + 1$ clearly labelling any intercepts with the coordinate axes and other important features.

3

- 6 -

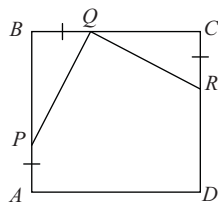
Question 9 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the exact area of an equilateral triangle with side length 4 centimetres. 2



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- (b) In the diagram, $ABCD$ is a square. The points P , Q , and R lie on AB , BC , and CD respectively such that $AP = BQ = CR$.



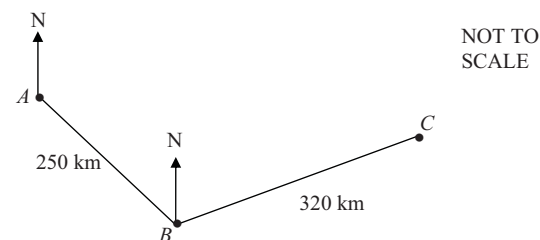
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Copy or trace the diagram into your writing booklet.

- (i) Prove that $\triangle PBQ \cong \triangle QCR$. 3
- (ii) Prove that $\angle PQR$ is a right angle. 2
- (c) Solve $2\sin^2 x = \sin x$ for $0^\circ \leq x \leq 360^\circ$. 2
- (d) Let α and β be the roots of the quadratic equation $4 - 8x - x^2 = 0$.
- (i) Find $\alpha + \beta$. 1
- (ii) Find $\alpha\beta$. 1
- (iii) Find $\alpha^2 + \beta^2$. 2
- (iv) Find $(\alpha - \beta)^2$. 2

Question 10 (15 marks) Use a SEPARATE writing booklet.

- (a) Solve $4 + 23x^2 - 6x^4 = 0$. 2
- (b) Determine whether the function $f(x) = 2x - x^3$ is even, odd or neither. 2
- (c) Write x^2 in the form $A(2-x)^2 + B(2-x) + C$. 3
- (d) A ship sailed 250 kilometres from Port A on a bearing of 153° and arrived at Port B to pick up some passengers. It then progressed 320 kilometres to Port C on a bearing of 071° .



Copy or trace this diagram into your writing booklet.

- (i) Find the distance AC correct to the nearest kilometre. 3
- (ii) Hence, find the bearing of Port A from Port C correct to the nearest degree. 2
- (e) Find $\frac{d}{dx} \left(\frac{4x^2}{\sqrt{3x-2}} \right)$. 3

Write your answer in simplest form with positive integer indices.

End of Paper

Examination Number:

Set:

Mathematics 2 Unit

Section I – Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct ↗

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

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1. D 4. D
2. C 5. B
3. A

Question 6

(a) Original Price = $\frac{\$1056}{1.1}$
= $\underline{\underline{\$960}}$

(b) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
= $\frac{|4(-2) + 3(5) - 12|}{\sqrt{4^2 + 3^2}}$
= $\frac{|-5|}{5}$
= 1 unit

(c) $x^2 - y^2 + 2x + 2y$
= $(x-y)(x+y) + 2(x+y)$
= $(x+y)(x-y+2)$

(d) $|3x-1| < 7$
 $-7 < 3x-1 < 7$
 $-6 < 3x < 8$
 $\underline{\underline{-2 < x < 2\frac{2}{3}}}$

(e) $-\frac{6}{m} = \frac{-3}{2}$
 $-12 = -3m$
 $\underline{\underline{m = 4}}$

(f) $(n-2) \times 180^\circ = 3960^\circ$
 $n-2 = 22$
 $\underline{\underline{n = 24}}$

(g) $3x^2 - 4x + 1 = 0$
 $(3x-1)(x-1) = 0$
 $x = \frac{1}{3} \text{ or } x = 1$

Question 7

(a) $\frac{d}{dx} (1-4x^2)^5$
= $5(1-4x^2)^4 \times -8x$
= $\underline{\underline{-40x(1-4x^2)^4}}$

(b) (i) $M = \left(\frac{-8+4}{2}, \frac{5+3}{2}\right)$
= $\underline{\underline{(-2, 1)}}$

(ii) $m_{AB} = \frac{-3-5}{4+8}$
= $\frac{-8}{12}$
= $\underline{\underline{-\frac{2}{3}}}$

(iii) $m_{AC} = \frac{3}{2}$ $M(-2, 1)$
 $y-1 = \frac{3}{2}(x+2)$
 $2y-2 = 3x+6$
 $\underline{\underline{3x-2y+8=0}}$

(iv) $5x + y + 35 = 0$ (1)
 $3x - 2y + 8 = 0$ (2) } Solve:
 $10x + 2y + 70 = 0$ (1e)
 $3x - 2y + 8 = 0$ (2)
(1e) + (2) $13x + 78 = 0$

$x = -6$
Sub $x = -6$ into (1)
 $5(-6) + y + 35 = 0$
 $y + 5 = 0$

$y = -5$
 $\therefore C \text{ is } \underline{\underline{(-6, -5)}}$

(v) $m_{AC} = \frac{-5-5}{-6+8} = \frac{-10}{2} = -5$
 $m_{BC} = \frac{-5+3}{-6-4} = \frac{-2}{-10} = \frac{1}{5}$

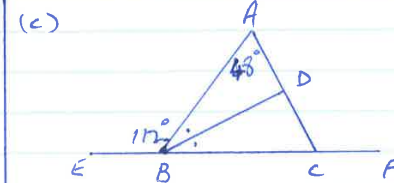
$m_{AC} \times m_{BC} = -5 \times \frac{1}{5} = -1$

$\therefore \underline{\underline{AC \perp BC}}$

(vi) $AC = \sqrt{(-8+6)^2 + (5+5)^2}$
= $\sqrt{4+100}$
= $\sqrt{104}$

$BC = \sqrt{(4+6)^2 + (-5+3)^2}$
= $\sqrt{100+4}$
= $\sqrt{104}$

$\therefore \text{Area} = \frac{1}{2} \times \sqrt{104} \times \sqrt{104}$
= $\underline{\underline{52 \text{ units}^2}}$



$\angle ABC = 68^\circ$ straight $\angle EBC$
 $\angle ABD = 34^\circ$ BD bisects $\angle ABC$
 $\angle BDC = 82^\circ$ Sum of 2 interior $\angle$ s equals opp. ext. $\angle$

Question 8

(a) $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sin \theta}{\sqrt{1-\sin^2 \theta}}$
= $\frac{\sin \theta}{\cos \theta} \times \sqrt{\cos^2 \theta}$
= $\frac{\sin \theta}{\cos \theta} \times \cos \theta$
= $\underline{\underline{\sin \theta}}$

(b) $f(x) = \frac{1}{2} x^{-4}$
 $f'(x) = -2x^{-5}$
= $\underline{\underline{-\frac{2}{x^5}}}$

(c) $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{2x + 8}$
= $\lim_{x \rightarrow -4} \frac{(x+4)(x-1)}{2(x+4)}$
= $\underline{\underline{-\frac{5}{2}}}$

$$(d) \quad y = x^3 - 3x^2$$

$$y' = 3x^2 - 6x$$

$$\text{at } x = -1 \quad y = (-1)^3 - 3(-1)^2$$

$$= -1 - 3$$

$$= -4$$

$$m_T = 3(-1)^2 - 6(-1)$$

$$= 3 + 6$$

$$= 9$$

$$\therefore \text{equation is: } y + 4 = 9(x + 1)$$

$$y + 4 = 9x + 9$$

$$y = 9x + 5$$

$$(e) \quad \text{For real roots } \Delta \geq 0$$

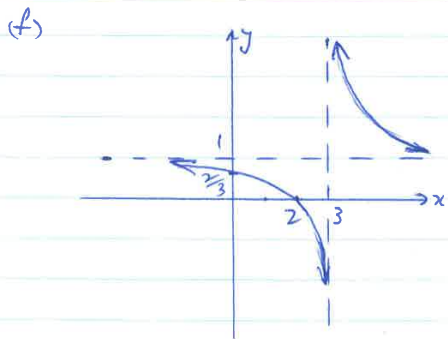
$$(-2k)^2 - 4 \times 3 \times k \geq 0$$

$$4k^2 - 12k \geq 0$$

$$4k(k - 3) \geq 0$$

$$\begin{array}{c} \vee \quad \times \quad \vee \\ 0 \quad \quad 3 \end{array}$$

$$\therefore \underline{k \leq 0 \text{ or } k \geq 3}$$

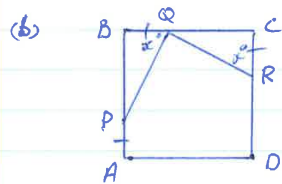


Question 9

$$(a) \quad \text{Area} = \frac{1}{2} \times 4 \times 4 \times \sin 60^\circ$$

$$= 8 \times \frac{\sqrt{3}}{2}$$

$$= 4\sqrt{3} \text{ units}^2$$



$$(i) \quad \angle B = \angle C = 90^\circ \text{ angles in square}$$

$$BQ = CR \text{ given}$$

$$BP = QC \text{ subtraction of equal length from sides of square}$$

$$\therefore \underline{\triangle PBQ \cong \triangle QCR \text{ SAS}}$$

$$(ii) \quad \angle BQP = \angle QRC = x^\circ \text{ corresp. } \angle \text{s in congruent } \triangle \text{s,}$$

$$\angle CQR = 90^\circ - x^\circ \text{ } \angle \text{ in } \triangle QCR$$

$$\angle PQR = 180^\circ - (x^\circ + 90^\circ - x^\circ) \text{ st. } \angle \text{ BQC}$$

$$= 90^\circ$$

$$(c) \quad 2\sin^2 x = \sin x$$

$$2\sin^2 x - \sin x = 0$$

$$\sin x (2\sin x - 1) = 0$$

$$\sin x = 0 \quad \sin x = \frac{1}{2}$$

$$\underline{x = 0^\circ, 180^\circ, 360^\circ} \text{ or } \underline{x = 30^\circ, 150^\circ}$$

$$(d) \quad (i) \quad \alpha + \beta = -\frac{b}{a}$$

$$= -\frac{-8}{-1}$$

$$= -8$$

$$(ii) \quad \alpha\beta = \frac{c}{a}$$

$$= \frac{4}{-1}$$

$$= -4$$

$$(iii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (-8)^2 - 2(-4)$$

$$= 72$$

$$(iv) \quad (\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$$

$$= 72 - 2(-4)$$

$$= 80$$

Question 10

$$(a) \quad 4 + 23x^2 - 6x^4 = 0$$

$$(4 - x^2)(1 + 6x^2) = 0$$

$$4 - x^2 = 0 \quad 1 + 6x^2 = 0$$

$$x^2 = 4 \quad x^2 = -\frac{1}{6}$$

$$x = \pm 2 \quad \text{no solution}$$

$$(b) \quad f(x) = 2x - x^3$$

$$f(-x) = 2(-x) - (-x)^3$$

$$= -2x + x^3$$

$$= -(2x - x^3)$$

$$= -f(x)$$

$$f(-x) = -f(x) \therefore \underline{f(x) \text{ is odd}}$$

$$(c) \quad x^2 \equiv A(2-x)^2 + B(2-x) + C$$

$$= A(4 - 4x + x^2) + 2B - Bx + C$$

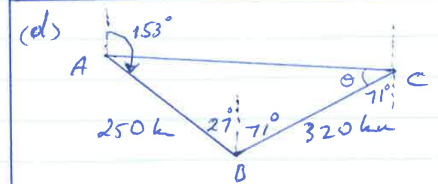
$$= Ax^2 + (-4A - B)x + (4A + 2B + C)$$

$$\underline{A=1} \quad -4A - B = 0 \quad 4A + 2B + C = 0$$

$$-4 - B = 0 \quad 4 - 8 + C = 0$$

$$\underline{B=-4} \quad \underline{C=4}$$

$$\therefore \underline{x^2 \equiv (2-x)^2 - 4(2-x) + 4}$$



$$(i) \quad AC^2 = 250^2 + 320^2 - 2 \times 250 \times 320 \cos 98^\circ$$

$$= 187167.6962$$

$$\underline{AC \doteq 433 \text{ km}}$$

$$(ii) \quad \frac{\sin \theta}{250} = \frac{\sin 98^\circ}{433}$$

$$\sin \theta = \frac{\sin 98^\circ}{433} \times 250$$

$$\sin \theta = 0.5717 \dots$$

$$\underline{\theta \doteq 35^\circ}$$

$$\therefore \text{bearing} = 190^\circ + 71^\circ + 35^\circ$$

$$= \underline{296^\circ}$$

$$(e) \quad \frac{d}{dx} \left(\frac{4x^2}{\sqrt{3x-2}} \right) = \frac{(3x-2)^{\frac{1}{2}} \cdot 8x - 4x^2 \cdot \frac{1}{2}(3x-2)^{-\frac{1}{2}}}{3x-2}$$

$$= \frac{(3x-2) \cdot 8x - 6x^2}{(3x-2)^{\frac{3}{2}}(3x-2)}$$

$$= \underline{\underline{\frac{18x^2 - 16x}{(3x-2)^{\frac{3}{2}}}}}$$